Name…….. Stream ……..

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE–REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

THE CITY COUNCIL OF DODOMA (CCD)

FORM ONE TERMINAL EXAMINATION 2022

031-PHYSICS

TIME:2:30Hours **JULY,2022**

INSTRUTIONS

1. This paper consists of sections **A**, **B** and **C** with a total of ten (**10**)questions
2. Answer all questions in all sections. Show your work clearly
3. All answers must be written in the spaces provided
4. All writing must be in blue or black ink except drawings which must be in pencil.
5. All communication devices, calculators and any unauthorized materials are not allowed in the examination room.
6. Write your Name and Stream at the top of every page.
7. Use the following if necessary:
   Force of gravity, g = 10 N/kg,
   Density of water = 1g/cm^3 or
   1000kg/m^3 Use, $\pi = 3.14$

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SECTION A (15 MARLKS)

1. Choose the best answer among the items given below questions.

(i) What is the reading in seconds indicated on the digital stopwatch shown in the figure below



A. 106s B. 146.78s C. 26s D. 224s

(ii) What is the name given to the people who study and work professionally in the field which relates matter and energy?
A. Scientists B. Physicians C. Physicists D. Meteorologists

(iii) Which of the following SI Units describes the fundamental quantity
A. m^2 B. N C. m D. kg/m^3

(iv) Archimedes principle states that:-
A. The weight acting on an object is equal to the volume of liquid displaced
B. The buoyant force acting on an object equals the weight of fluid displaced
C. The buoyant force acting on an object will always be equal to the up thrust
D. The weight of the liquid displaced is equal to weight of an object.

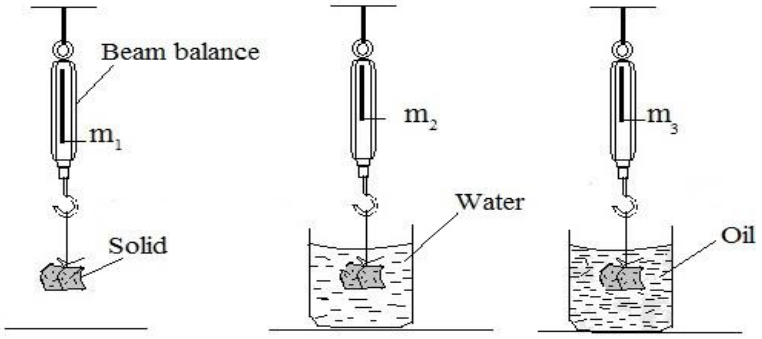
(v) What differentiate science from technology
A. Science involve study of universe, technology is the application of science
B. Technology is the application of nature, science is physics, chemistry and biology
C. Science study of matter, technology is the application of science
D. Technology is the application of science, science is the study of nature

(vi) The following observations are correct; except
A. You should move the victim of electric shock using a metallic object
B. First AID helps to preserve life
C. Eating in the laboratory is prohibited
D. Thermometer are used for measuring body temperature

(vii) A student is told to measure out exactly 25cm^3 of water. Which of the following apparatus is the best for this measurement?
A. Measuring cylinder B. Beaker C. Pipette D. Burette

(viii) A form one student at Karume secondary school wants to measure the volume of a stone with an irregular shape. Which of the following pair of instruments would you recommend her to use?
A. Measuring cylinder and beaker
B. Measuring cylinder and overflow can
C. Metre rule and overflow can
D. Metre rule and beaker

(ix) The diagram below shows series of experiments performed by a form one student to determine the relative density of the oil. Three reading of the beam balance were



obtained.

The relative density of the oil is given by;-

A. $\frac{m_1}{m_1 - m_2}$ B. $\frac{m_1 - m_2}{m_1 - m_3}$ C. $\frac{m_3 - m_1}{m_3 - m_2}$ D. $\frac{m_1 - m_3}{m_1 - m_2}$

(x) A body weighs 3.6N in air and 2.4N when full immersed in water. The density of the body in g/cm^{3}is
A. 0.40 B. 0.80 C. 2.00 D. 3.00

2. Match item in **LIST A** with responses in **LIST B** by writing the letter of the correct responses beside the item number.

Name …….. Stream ……..

LISTA	LIST B
(i) Law of buoyancy	A. Floating object
(ii) Law of submergence	B. Sinking object
(iii) Object has weight of 15N and a buoyant force of 18N	C. Buoyed up object
(iv) Upthrust less than weight of an object	D. Law of flotation
(v) Upthrust equal to the weight of an object	E. Archimedes principle

SECTION B (70 MARKS)

3. For each of the items (i) – (v) fill in the blanks by writing the correct answer.

(i) Are the guidelines to be followed in the laboratory in order to reduce risk of accidents……

(ii) Sets of technologies used by scientists to investigate a problems or answer a questions known as……

(iii) An instrument which can be used to measure the depth of the tea cup……

(iv) A solid ball has a mass of 333 g and radius of 2cm, the density of the solid ball is……..

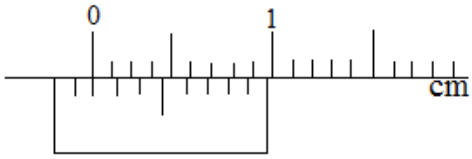
(v) Is the force which holds the constituents of the atomic nucleus……

4. (a) Mention two (2) features of a good laboratory;-

(b) What are the two steps you should take before administering First Aid to an injured person?

5. (a) Mention three (3) ways of reducing errors during measurement.

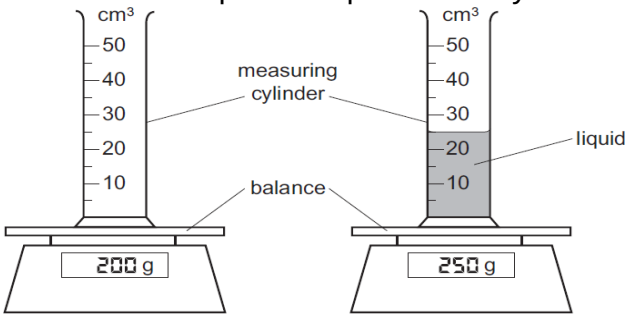
(b) The figure below shows part of a vernier caliper when the jaws are closed without an object in between the jaws.



A student used this vernier caliper to measure the diameter of a test tube of actual diameter 2.15cm.What was the reading shown by the vernier caliper?

6. (a) Density of iron is $7.8g/cm^3$, what does this statement mean?

(b) The diagram shows an experiment performed by a form one student to find the



density of a liquid. The following was the readings of measuring cylinder when was empty and when filled with the liquid.

What is the density of the liquid?

7. (a) Briefly explain why a spoon sinks but a steel ship floats
(b) When a body of weight 6.4N immersed in water it weighs 4.8 N. Therefore the upthrust acting on the body is …… and weight of water displaced is……By finding mass of water displaced we can find volume of water displaced which is ……Therefore volume of the object is ……and its density is ……

8. (a) (i) What is a Newton;-

(ii) Give a reason why it is more difficult to pull a boat on the beach than in the sea.

(b) An astronaut weighs 800N on earth. Determine his mass on planet Mars if the gravitational field strength of the planet Mars is 3.8N/kg.

9. (a) A stone weighs 560 g when completely immersed in kerosene whose density is 0.8 g/cm^3. If the mass of displaced kerosene is 320 g. Calculate:
(i) Volume of the stone

(i) Density of the stone.

(b) A boat floating on water has mass of 1000 kg and volume of 10m^3

(i) What is the upthrust acting on it

(ii) What percentage of the boat is above water?

SECTION C (15 Marks)

10 (a) With the aid of diagrams explain simple experiment using magnets to demonstrate **attractive** and **repulsive forces**

(b) Indicate the type of Physics apparatus which can be used in measurements of the following;-
(i) 2kg of sugar bought from Massawe shop ……
(ii) Diameter of a wire measured by a technician in electronics devices repairing shop……
(iii) The amount of electric current flowing in extension socket ……..
(iv) The size of your school uniform measured by a tailor ……
(c) Draw a diagram of which indicates the forces acting on a object immersed in a liquid